

From Neighborhood Thinning to Cosmic Expansion: Quantitative Tests of a Relational-Complexity Hypothesis

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Abstract

Adding genuinely distinguishing coordinates to an object increases its separation from alternatives and thins every fixed-radius neighborhood under a broad class of additive distances. The result invites a speculative but testable question: could physical expansion reflect the growth of relational distinguishability? We construct the simplest bridge by identifying inverse exact-match density with relational volume, obtaining $H = (\ln 2/3)\dot{N}$. A minimal history law then predicts $H_0 = 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1}$ from the measured cosmic age, placing the observed Hubble scale within a few percent without inserting H_0 . The coincidence is instructive but incomplete: the same law produces a coasting universe and is strongly rejected by the DESI DR2 baryon-acoustic-oscillation distance shape. A two-channel extension fits the BAO data well, but is algebraically identical to the flat- Λ CDM background. Replacing accumulated time by relativistic causal-past four-volume also fails to yield a constant expansion-complexity coupling. Finally, public CAMELS simulations show that neighborhood change is enhanced in dense, bound environments where expansion is suppressed. Most of this signal is predicted by peculiar motion, and it persists in matched gravity-only simulations. These results do not eliminate a relational account of spacetime. They narrow its viable content: the tested complexity measures behave more naturally as diagnostics of gravitational dependency and binding than as a local source of dark-energy expansion.

A geometric question with cosmological stakes

The discovery that the cosmic expansion is accelerating transformed the cosmological constant from a formal possibility into an empirical problem [1, 2]. Flat Λ CDM remains an economical and highly successful description of the background, particularly when confronted with the cosmic microwave background and baryon acoustic oscillations [3, 4]. Yet its dark-energy scale is entered as a fitted constant rather than derived from a microscopic principle. That gap has kept alive a broad question: might geometry be an effective description of deeper statistical, thermodynamic, or informational structure?

There is serious precedent for asking the question carefully. Einstein’s equation can be recovered as a local thermodynamic equation of state under specific horizon-entropy assumptions [5], while causal-set theory explores how Lorentzian order and spacetime volume might emerge

from a locally finite relational structure [6]. These programs do not license an arbitrary identification of information with geometry. They do, however, show what a successful bridge must eventually provide: a covariant object, a dynamical law, and quantitative contact with observation.

The present work begins from a much simpler exact fact. Under additive nonnegative distances, appending a distinguishing coordinate can only increase pairwise separation and can only reduce the population inside a fixed-radius neighborhood. For independent binary coordinates, exact-match density falls as 2^{-N} . If the inverse of this density is interpreted as an effective relational volume, then complexity growth has the kinematics of expansion.

Our purpose is not to promote that analogy by rhetoric. It is to give it every reasonable quantitative opportunity to succeed, then identify precisely what survives. We proceed in four stages: derive the exact kinematic bridge, test simple history laws against the Hubble scale and the DESI DR2 distance shape, replace naive accumulated time by a relativistic causal-history construction, and examine neighborhood evolution directly in hydrodynamic and gravity-only cosmological simulations.

The exact bridge—and the missing dynamics

Consider objects described by N coordinates, with additive distance

$$\delta_N(O_i, O_j) = \sum_{n=1}^N d_n(o_{in}, o_{jn}), \quad d_n \geq 0. \quad (1)$$

Appending a coordinate gives $\delta_{N+1} = \delta_N + d_{N+1} \geq \delta_N$. Consequently every fixed-radius neighborhood is nonincreasing with N . In the particularly transparent case of independent uniform binary coordinates and exact matching, the expected neighborhood density is

$$\rho_N = 2^{-N}. \quad (2)$$

Define an effective relational volume and a three-dimensional scale factor,

$$V_{\text{rel}} = \rho_N^{-1}, \quad a_{\text{rel}} = V_{\text{rel}}^{1/3} = 2^{N/3}. \quad (3)$$

The corresponding expansion rate is then exact:

$$H_{\text{rel}} \equiv \frac{\dot{a}_{\text{rel}}}{a_{\text{rel}}} = \frac{\ln 2}{3} \dot{N}. \quad (4)$$

Equation (4) is the central bridge, but also the central warning. It is a dictionary, not a field equation. The factor of three comes from the chosen volume interpretation; the bit rate $\dot{N}$ has not been derived; and no stress-energy tensor or conservation law has appeared. Physical content begins only when N is defined independently of the observed scale factor and its dynamics predict more than one number.

A numerical coincidence worth testing

Suppose that maintaining a distinguishable three-dimensional history requires one new binary distinction per spatial dimension whenever the temporal baseline doubles:

$$N(t) = 3 \log_2(t/t_*). \quad (5)$$

Equations (4) and (5) imply

$$a(t) \propto t, \quad H(t) = t^{-1}. \quad (6)$$

Using $t_0 = 13.80$ Gyr gives

$$H_0^{\text{age}} = 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (7)$$

This is 5.1% above the Planck base- Λ CDM value $67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [3], 3.9% above the DESI DR2+CMB value used here, and 3.0% below the SH0ES distance-ladder value $73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [7]. Because CMB-derived age and H_0 are correlated and model dependent, this is not a precision independent prediction. It is nevertheless a striking dimensional result: a transparent information law lands on the observed Hubble scale without inserting it.

The same simplicity makes the decisive failure immediate. Equation (6) gives $\ddot{a} = 0$ and $q = 0$, whereas the measured late universe accelerates. A correct value of H at one epoch is not a correct expansion history. The coincidence therefore earns a shape test rather than a claim.

DESI tests the history, not merely the scale

Baryon acoustic oscillations provide a standard-ruler measurement of transverse and radial cosmic distances [8]. We use the published 13-element DESI DR2 BAO data vector and covariance [4]. For every model, the nuisance normalization

$$X = \frac{c}{H_0 r_d} \quad (8)$$

is profiled analytically. The comparison therefore tests the redshift dependence of the distance relation without using the value of H_0 that motivated the hypothesis.

For the family

$$N(t) = d_{\text{eff}} \log_2(t/t_*), \quad a(t) \propto t^p, \quad p = d_{\text{eff}}/3, \quad (9)$$

we test a matter-like history, the coasting spatial history, a literal “3D plus time” history, and a freely fitted constant p . Table 1 and Fig. 1 show the result. No constant-dimensional history describes the BAO shape. Even the

Table 1: DESI DR2 BAO shape fits. The BAO normalization is fitted in every row. The two-channel row also fits its channel ratio.

Model	χ^2	dof
$p = 2/3$	1420.18	12
$p = 1$	187.09	12
$p = 4/3$	1364.93	12
Best constant $p = 0.9125$	96.97	11
Two channel	10.27	11
Flat Λ CDM, $\Omega_m = 0.3027$	10.63	12

best constant power gives $\chi^2 = 96.97$ for 11 degrees of freedom. More sharply, fitting p and X only below $z = 1$ gives $p = 1.1185$, but the six held-out measurements at $z \geq 1$ yield $\chi^2 = 670.72$ without refitting.

The successful extension and its honest interpretation

A more flexible relational construction separates local material interactions from a persistent history channel and combines statistically independent rates in quadrature:

$$\dot{N}_{\text{eff}}^2 = \dot{N}_{m,0}^2 a^{-3} + \dot{N}_h^2. \quad (10)$$

Together with Eq. (4),

$$H^2(a) = \left(\frac{\ln 2}{3} \right)^2 \left(\dot{N}_{m,0}^2 a^{-3} + \dot{N}_h^2 \right). \quad (11)$$

Fitting the channel ratio and BAO scale gives $\Omega_{m,\text{eff}} = 0.2975$ and $\chi^2 = 10.27$ for 11 degrees of freedom. A fit restricted to $z < 1$ gives $\Omega_{m,\text{eff}} = 0.3095$ and predicts the six high-redshift measurements with $\chi^2 = 4.74$. With the reference ruler $r_d = 147.09 \text{ Mpc}$, the full fit corresponds to $H_0 = 69.03 \text{ km s}^{-1} \text{ Mpc}^{-1}$, a present material-channel rate of $0.1667 \text{ bits Gyr}^{-1}$, and a history-channel rate of $0.2561 \text{ bits Gyr}^{-1}$.

This is a useful success of form, but not yet a new explanation. Equation (11) is exactly the flat- Λ CDM background equation under a change of names. It predicts the familiar acceleration transition, $z_{\text{acc}} \simeq 0.68$, but becomes explanatory only if the quadratic composition rule, the a^{-3} term, the constant rate, and the approximately 0.30:0.70 channel ratio follow from an independent microscopic theory. The fit tells us what must be derived; it does not perform the derivation.

Time must be made relativistic

The literal addition of time as a fourth positive metadata coordinate is not covariant. Cosmic time is not an interchangeable Euclidean feature, and relativistic observables are organized by causal structure. Motivated by the role of causal order in discrete approaches to spacetime [6], we replace accumulated coordinate time by the causal past of an observation event. In flat FLRW spacetime its

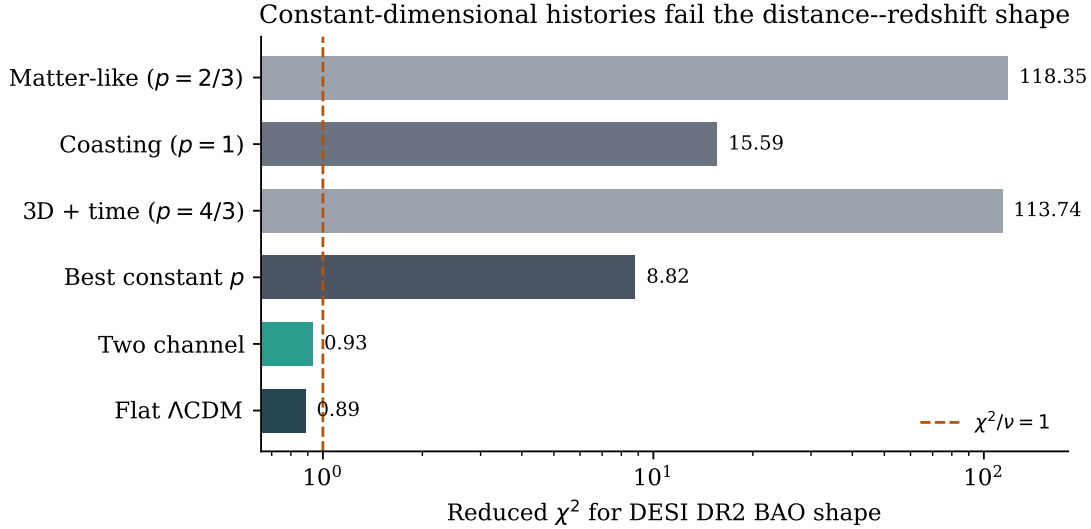


Figure 1: Reduced χ^2 for the DESI DR2 BAO distance shape after profiling the common ruler normalization. The logarithmic axis makes visible both the strong rejection of constant-dimensional histories and the good background fit of the two-channel and flat- Λ CDM forms.

four-volume is

$$V_4(\tau_o) = \frac{4\pi}{3} \int_0^{\tau_o} d\tau' a^3(\tau') \left[\int_{\tau'}^{\tau_o} \frac{c d\tau''}{a(\tau'')} \right]^3. \quad (12)$$

The covariant expansion scalar is $\theta = \nabla_\mu u^\mu = 3H$. We ask whether a redshift-independent constitutive relation

$$3H = \kappa \dot{C} \quad (13)$$

can hold for either $C = V_4$ in fixed units or $C = \log_2 V_4$. Evaluating Eq. (12) from $z = 2.33$ to the present along the DESI-calibrated two-channel background, the required κ varies by a factor of 321.8 for raw four-volume. Logarithmic volume performs substantially better, but still requires a factor 1.514 drift. Its present coupling is 0.5373, rather than the $\ln 2 = 0.6931$ supplied by the direct neighborhood-volume dictionary.

This result is more constructive than simply declaring “time is different.” It identifies the problem: total causal volume repeatedly counts shared and redundant history. If a history channel exists, the more plausible object is the *conditional new information* between neighboring causal slices after conditioning on their common past. That object is not supplied by Eq. (12), so we next seek an empirical analogue in evolving cosmological neighborhoods.

Neighborhood evolution in CAMELS

CAMELS provides many $25 h^{-1} \text{Mpc}$ cosmological volumes, including IllustrisTNG hydrodynamic simulations and gravity-only counterparts [9, 10]. The hydrodynamic runs use the IllustrisTNG galaxy-formation framework [11]. These simulations assume an FLRW/ Λ CDM background and therefore cannot establish that information

causes cosmic expansion. They can test the narrower local prediction: does independently measured neighborhood novelty rise where physical expansion is stronger, and is it suppressed in bound regions?

Tracking and estimators

We analyze snapshots 86, 88, and 90, corresponding approximately to $z = 0.100, 0.049$, and 0. Subhalos with at least 100 particles are matched between adjacent snapshots using mutual-nearest comoving positions, a mass ratio within a factor of three, and a maximum displacement of $0.5 \text{ Mpc}/h$. Where the most-bound particle remains available, this geometric procedure recovers the persistent identity in 94–98% of cases across the analyzed samples.

For every matched object we construct a $k = 16$ nearest-neighbor set. A discrete estimator assigns surprisal to newly appearing neighbors conditional on the old set. A continuous robustness estimator assigns Gaussian weights over the union of 64 nearby candidates and measures the Jensen–Shannon divergence between successive neighborhood distributions [12]. The physical volume change follows the same old neighbors into the next snapshot, using the determinant of their local covariance ellipsoid and restoring the background a^3 factor. Low-density centrals define the field sample. Satellites or objects in the upper local-density quartile define a bound-region proxy.

In the initial fixed-cosmology volume, neighbor retention is 96.5–96.6%. The field follows global expansion, with local-to-global ratios 0.982 and 0.989, while the bound proxy gives 0.460 and 0.455. Yet the continuous neighborhood JS change is roughly 24 times larger in the bound proxy. Novelty correlates positively with density and negatively with local expansion. The same environmental separation persists for $k = 8, 32$, and 64. Thus

the direct prediction has the wrong local sign: the largest relational rearrangements occur not in freely expanding neighborhoods but in structured regions of orbital motion, close passages, and mergers.

Conditioning on phase space

Much of this rearrangement can be reversible dynamics rather than new information. We therefore propagate each old phase-space state using its measured peculiar velocity. In comoving Mpc/h, the first-order FLRW prediction is

$$\mathbf{x}_{\text{pred}}(a_2) = \mathbf{x}(a_1) + \frac{\mathbf{v}_{\text{pec}}(a_1)}{100} \int_{a_1}^{a_2} \frac{da}{a^2 E(a)}, \quad (14)$$

$$E(a) = \sqrt{\Omega_m a^{-3} + \Omega_\Lambda}.$$

The residual is the JS divergence between the neighborhood predicted by Eq. (14) and the observed neighborhood. This is a deliberately conservative ballistic control: accelerations, mergers, disruption, and halo identification remain in the residual.

Across five independent fixed-cosmology IllustrisTNG realizations and two intervals per realization, field patches expand at 0.992 ± 0.021 of the global rate, whereas the bound proxy expands at 0.500 ± 0.089 . Raw median JS changes are $(3.80 \pm 0.85) \times 10^{-5}$ bits in the field and $(8.39 \pm 1.53) \times 10^{-4}$ bits in the bound proxy. Phase conditioning reduces these to $(1.33 \pm 0.32) \times 10^{-6}$ and $(1.40 \pm 0.64) \times 10^{-4}$ bits: reductions of 96.5% and 83.3%, respectively. The residual remains anticorrelated with expansion ($\rho = -0.304 \pm 0.047$) and correlated with density ($\rho = +0.629 \pm 0.032$). Quoted dispersions are sample standard deviations across realization–interval measurements, not cosmological parameter errors.

A matched gravity-only control

To isolate baryonic structural complexity, we repeat the analysis for matched IllustrisTNG and IllustrisTNG-DM Latin-hypercube realizations LH_0–LH_2, which share simulation indices, cosmologies, and initial seeds. The hydrodynamic and gravity-only samples yield field expansion ratios 1.000 ± 0.011 and 0.988 ± 0.017 , and bound ratios 0.533 ± 0.062 and 0.542 ± 0.049 . Their raw field JS changes are nearly identical. The raw bound change is $(1.041 \pm 0.046) \times 10^{-3}$ bits in hydrodynamics and $(1.241 \pm 0.116) \times 10^{-3}$ bits in gravity only; after phase conditioning the corresponding residuals are $(2.32 \pm 1.34) \times 10^{-4}$ and $(3.21 \pm 0.56) \times 10^{-4}$ bits (Fig. 2).

The qualitative effect therefore does not require gas physics, star formation, or feedback. Collisionless gravity is sufficient. This does not make the relational pattern uninteresting. It changes its most natural interpretation: neighborhood complexity records the dynamically maintained dependencies of bound structure rather than acting as an independent pressure that expands space.

What the tests leave standing

The original intuition separates into three claims that should no longer be conflated.

First, *structural distinction thins abstract neighborhoods*. This is an exact combinatorial statement under the stated distance assumptions, and none of the cosmological tests alters it.

Second, *inverse neighborhood density can be written as an effective volume*. This produces the exact kinematic relation Eq. (4). It is mathematically coherent and explains why the Hubble scale emerges so readily from a bit rate. But without independent dynamics, any measured $H(t)$ can be translated into a fitted $N(t)$, making the construction tautological.

Third, *complexity growth physically drives cosmic expansion*. The specific realizations tested here do not support this claim. Fixed-dimensional history laws fail the BAO shape; total causal history lacks a constant coupling; and direct local novelty is largest where expansion is smallest. A successful future model must introduce a covariant, coarse-grained, irreversible information functional and derive its gravitational coupling before using expansion data. It must also distinguish itself from Λ CDM in growth, lensing, redshift-space distortions, or another observable, rather than only relabeling the background equation.

The binding result offers a productive direction. In standard gravity, the competition between a bound mass M and late-time accelerated expansion sets a maximum turnaround scale

$$R_{\text{TA,max}} = \left(\frac{3GM}{\Lambda c^2} \right)^{1/3}, \quad (15)$$

which has been proposed as a test of Λ CDM and alternatives [13]. The CAMELS results suggest that relational dependency measures may illuminate the structured side of this boundary. The challenge is to derive—not rename—the dependence on G , M , and R , and to show whether an information balance reproduces the turnaround normalization.

Conclusion

The neighborhood-thinning idea survives contact with cosmology, but in a more interesting and disciplined form than the opening analogy. Its exact combinatorics provide a clean bridge from distinguishing information to an effective expansion rate. A minimal history law then lands unexpectedly close to the observed Hubble scale. Precision distance data show that the same law cannot describe the expansion history, while a successful two-channel version reveals itself as the standard flat- Λ CDM background in relational language. Making history relativistic improves the formulation but not yet the dynamics.

The simulations supply the most concrete new clue. Relational rearrangement is enhanced where gravity binds and cosmic expansion is suppressed; ordinary phase-space

■ Raw change ■ After velocity conditioning
 Neighborhood change concentrates in bound regions and is largely kinematic

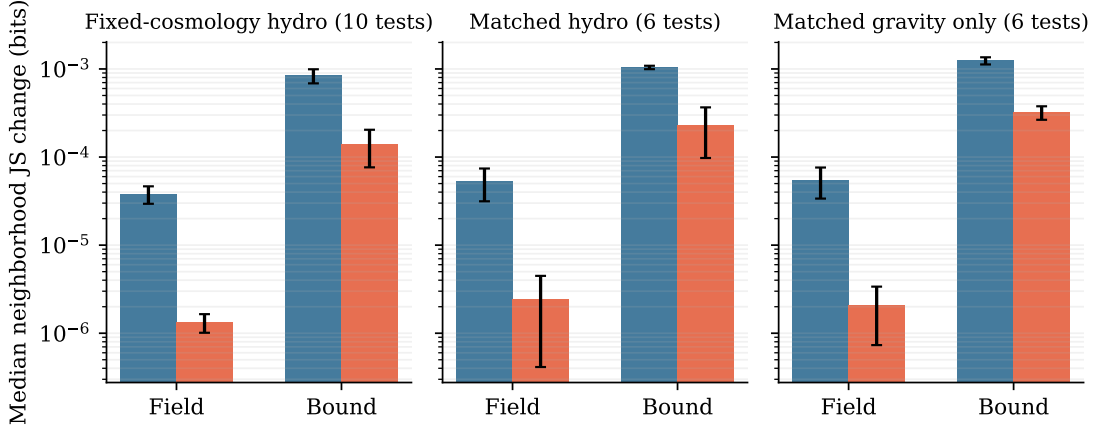


Figure 2: Continuous neighborhood change in field and bound-proxy samples. Velocity propagation removes most of the raw change, especially in the field. The bound excess and its survival in gravity-only simulations indicate that ordinary nonlinear gravitational dynamics, not baryonic complexity, is the dominant source of the measured signal. Error bars show sample standard deviations across realization–interval measurements.

motion explains most of it; and gravity alone reproduces it. The result redirects the hypothesis rather than ending it. Neighborhood complexity appears better suited to describing how gravitational structures maintain dependent histories than to serving, in its present form, as the source of dark energy. That narrowed target is now precise: construct an irreversible covariant information measure, derive its coupling, and demand a prediction that cannot be obtained by translating the observed Friedmann equation.

Data and reproducibility

All calculations are implemented in the accompanying repository. The DESI test uses the public DR2 BAO mean vector and covariance. CAMELS group catalogs are downloaded from the public release [10]; positions and velocities follow the published Illustris catalog conventions. The scripts record the selections, matching cuts, estimators, phase propagation, and all reported fit statistics. No CAMELS result is used as an independent test of the assumed background cosmology.

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